

Funding Proposal: GPU Compute Node for the Dartmouth Research Cluster

Summary

We propose a one-time investment of \$185,000 to purchase a high-end GPU compute node and add it to the existing Dartmouth research cluster (Slurm-scheduled, InfiniBand-connected). The node provides four NVIDIA H200 accelerators with 564 GB of pooled high-bandwidth memory. It addresses a gap in large-memory accelerated computing that currently sends AI and machine-learning research to metered cloud GPUs. As a shared cluster resource, it would serve multiple labs while supporting model training, fine-tuning, and high-throughput inference, and would help keep Dartmouth competitive in AI research.

Justification

- **Capability gap.** Training, fine-tuning, and serving current open models (roughly 70B to 400B parameters) need 100 GB or more of GPU memory and NVLink-coupled accelerators. The cluster has no large-VRAM GPUs today, so this work either waits in queue or runs on costly on-demand cloud instances.
- **Cost relative to cloud.** A comparable four-GPU H200 cloud instance rents for about \$14 to \$16 per hour, or roughly \$90,000 to \$120,000 per year at sustained research use. The node reaches break-even in about 18 to 24 months and then provides two to three further years of compute at only power and maintenance cost, on campus, with no data leaving the institution.
- **Competitiveness in AI research.** Large-memory GPUs are now standard infrastructure for competitive AI and machine-learning research. On-site capacity lets faculty and students train and fine-tune models and run inference at scale, and supports the externally funded AI projects that increasingly assume access to this class of hardware. It strengthens Dartmouth’s position for faculty recruiting, grant competitiveness, and collaborations.
- **Shared resource.** The node joins the existing Slurm scheduler, so idle cycles are available to other groups across campus.

Proposed System and Budget

Component	Specification	Cost (USD)
GPU accelerators	4 × NVIDIA H200 SXM, 141 GB HBM3e (HGX baseboard, NVLink/NVSwitch); 564 GB pooled	\$128,000
Server platform	Dual AMD EPYC (192 cores), HGX carrier, redundant 3 kW PSUs	\$20,000
System memory	1.5 TB DDR5 ECC	\$10,000
Local storage	About 30 TB NVMe (RAID1 OS plus high-speed scratch)	\$8,000
Cluster networking	NVIDIA ConnectX-7 NDR 400 Gb InfiniBand HCA plus cabling	\$6,000
Integration	Rack, rails, PDU, Slurm integration, burn-in testing	\$3,000
Warranty and support	3-year 24/7 on-site, next-business-day parts	\$10,000
Total		\$185,000

Cost figures are planning estimates from our June 2026 conversation with Dartmouth Research Computing (ITC) and are subject to final vendor quotes. Power and cooling are assumed available in the existing cluster facility.

Alternative configuration: for workloads dominated by many-user inference rather than large-model training, the same budget buys 8 × NVIDIA L40S (48 GB) for higher aggregate throughput. The H200 build above is recommended as the more training-capable option.

Expected Outcomes

- Continuous on-campus training and inference for AI and machine-learning research, without queueing for scarce accelerators.
- Reduced recurring cloud GPU spend once the node is in sustained use.
- Capacity to train and fine-tune models on institutional data, kept on campus.
- A shared accelerator resource that strengthens Dartmouth’s competitiveness for AI research, recruiting, and grants.